



Notes

1.Alert Priority Levels

Introduction

In a Security Operations Center (SOC), thousands of alerts are generated daily. Alert prioritization helps analysts identify which threats need immediate attention.

Priority Definitions:

SOC teams generally use four priority levels:

Priority Level	Description	Example
Critical	Immediate threat causing major damage or active attack	Ransomware encrypting files
High	Serious security issue needing urgent response	Unauthorized admin login
Medium	Suspicious activity with moderate impact	Multiple failed login attempts
Low	Minor or informational alerts	Antivirus update notification

1.Critical Alert

A critical alert indicates a severe security incident that can heavily impact the organization.

Examples

- Ransomware attack
- Domain admin compromise
- Database breach

Response Time

Immediate action required.

2. High Alert

A high alert represents a dangerous activity that may become critical if ignored.

Examples

- Unauthorized admin access
- Brute force attack
- Remote code execution attempt

3.Medium Alert



A medium alert indicates suspicious or unusual activity with limited impact.

Examples

- Repeated failed logins
- Suspicious PowerShell execution
- Unusual outbound traffic

4. Low Alert

Low alerts are informational or low-risk activities.

Examples

- User password reset
- Antivirus scan completed
- Normal port scan inside lab

Factors Used for Alert Prioritization

1.Asset Criticality

The importance of the affected system.

Asset Type	Priority Impact
Production Server	High
Domain Controller	Critical
Test VM	Low
Employee Laptop	Medium

2.Exploit Likelihood

Probability that attackers can exploit the vulnerability.

Example:

Log4Shell had public exploits available, so exploit likelihood was very high.

3. Business Impact

Measures damage to organization operations.

Example:

Attack on banking payment system = Critical business impact.



CVSS (Common Vulnerability Scoring System)

CVSS is an industry standard used to measure vulnerability severity.
Developed by FIRST (Forum of Incident Response and Security Teams).

Score	Severity
0.0	None
0.1 – 3.9	Low
4.0 – 6.9	Medium
7.0 – 8.9	High
9.0 – 10.0	Critical

CVSS Metrics

1. Base Metrics

Measures core characteristics of vulnerability.

Includes:

- Attack Vector
- Attack Complexity
- Privileges Required

2. Temporal Metrics

Measures factors changing over time.

Includes:

- Exploit availability
- Patch availability
- Report confidence

3. Environmental Metrics

Measures impact in specific organization environment.

Includes:

- Asset importance
- Security controls
- Business requirements

Log4Shell Example (CVE-2021-44228)

Overview

Log4Shell was a critical vulnerability in Apache Log4j.

CVSS Score



9.8 (Critical)

Why Critical?

- Remote code execution
- Public exploit available
- Affected many organizations

2. Incident Classification

A. Incident Categories

Security incidents are classified based on attack type and impact.

- **Malware** – Viruses, ransomware, spyware infections
- **Phishing** – Fake emails/websites used to steal credentials
- **DDoS** – Flooding a server/network to make services unavailable
- **Insider Threat** – Unauthorized actions by employees or trusted users
- **Data Exfiltration** – Stealing sensitive data from systems

Example:

Employee copies company files to personal USB → *Insider Threat*

B. Taxonomy Frameworks

Standard frameworks help analysts label incidents consistently.

- MITRE ATT&CK
Maps attacker tactics and techniques
Example: **T1566 – Phishing**
- ENISA Incident Taxonomy
Provides structured incident classification categories
- VERIS
Used for recording and sharing incident details

C. Contextual Metadata

Additional information added to incidents for better investigation.

- Affected systems
- Date and time of attack
- Source IP address



- Indicators of Compromise (IOCs)
 - Malicious file hash
 - Suspicious domain
 - Attack URL

3. Basic Incident Response

A. Incident Response Lifecycle

Preparation: This phase focuses on getting ready before an incident happens.

Activities:

Create IR policies and playbooks

Train security teams

Install security tools (SIEM, EDR, Antivirus)

Identification: Detect and confirm whether a security incident has occurred.

Activities:

- Monitor alerts and logs
- Perform alert triage
- Analyze suspicious activity

Containment: Stop the attack from spreading.

Activities:

- Isolate infected systems
- Disable compromised accounts
- Block malicious IPs/domains

Eradication: Remove the root cause of the attack.

Activities:

- Delete malware
- Remove malicious files
- Patch vulnerabilities

Recovery



Restore systems and services safely.

Activities:

- Restore backups
- Monitor systems for reinfection
- Return systems to production

Important Incident Response Procedures

1.System Isolation

Separating infected devices from the network to stop spread.

2. Evidence Preservation

Collecting and protecting evidence for investigation.

Methods:

- Memory dumps
- Log collection
- File hashing

3.Communication Protocols

Proper communication during incidents.

Includes:

- Informing management
- Reporting to SOC team
- Escalation procedures

4.SOAR (Security Orchestration, Automation, and Response)

Tools that automate incident response tasks.

Functions:

- Automate alert handling
- Create workflows
- Reduce manual work



1.Alert Management Practice

Objective

Alert classification, prioritization, dashboard creation, incident ticketing, and escalation handling.

Tools Used:

Wazuh

Thehive

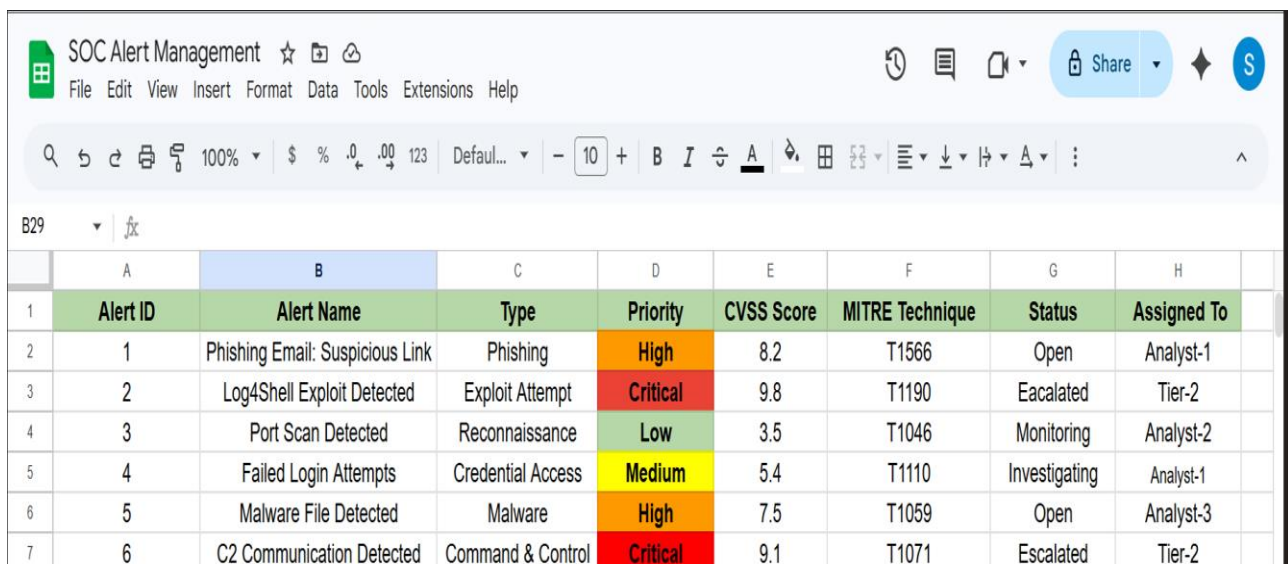
Google sheet

Task1:Alert Classification System:

Map alerts to MITRE ATT&CK techniques and classify alerts based on severity

Steps:

1. Open Google Sheets
- 2.create Columns
3. Alert ID
- 4.Alert Type
- 5.Priority
- 6.MITRE Technique Description
- 7.Add mock alerts.



SOC Alert Management								
File Edit View Insert Format Data Tools Extensions Help								
100% 123 Default... 10 B I A								
B29 fx								
	A	B	C	D	E	F	G	H
1	Alert ID	Alert Name	Type	Priority	CVSS Score	MITRE Technique	Status	Assigned To
2	1	Phishing Email: Suspicious Link	Phishing	High	8.2	T1566	Open	Analyst-1
3	2	Log4Shell Exploit Detected	Exploit Attempt	Critical	9.8	T1190	Eacalated	Tier-2
4	3	Port Scan Detected	Reconnaissance	Low	3.5	T1046	Monitoring	Analyst-2
5	4	Failed Login Attempts	Credential Access	Medium	5.4	T1110	Investigating	Analyst-1
6	5	Malware File Detected	Malware	High	7.5	T1059	Open	Analyst-3
7	6	C2 Communication Detected	Command & Control	Critical	9.1	T1071	Escalated	Tier-2



Task2: Alert Prioritization Using CVSS

Prioritize alerts based on CVSS score.

8				
9	CVSS Priority Guide			
10	CVSS Score Range	CVSS Score Range	Description	
11	9.0-10.0	Critical	Urgent: Immediate action required	
12	7.0-8.9	High	High risk: Investigate and respond	
13	4.0-6.9	Medium	Moderate risk: Monitor and analyze	
14	0.1 – 3.9	Low	Low risk: Monitor	
15				
16				

Priority Logic

- Critical → Immediate escalation
- High → Investigate within 1 hour
- Medium → Investigate during shift
- Low → Monitor only

Procedure:

- 1.Create a new sheet named “CVSS Prioritization”.
- 2.Add CVSS score column.
- 3.Apply color coding:
- 4.Red = Critical
- 5.Orange = High
- 6.Yellow = Medium
- 7.Green = Low



Task 3 – Dashboard Creation in Wazuh

Objective:

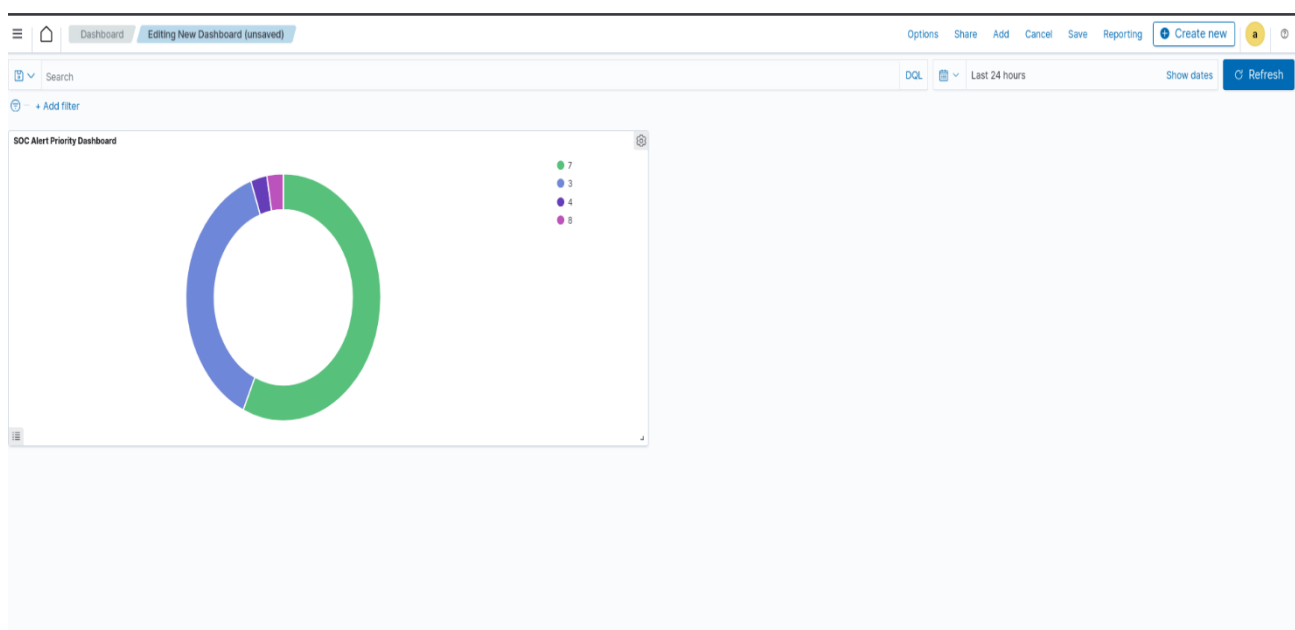
Visualize alert priorities using charts

Steps:

1. Login to Wazuh Dashboard.
2. Open: Dashboard →
3. Create Dashboard Add Visualization:
4. Pie Chart or Donut Chart
5. Use fields:
6. rule.level
7. rule.description
8. Save dashboard with name:
9. SOC Alert Priorities

Recommended Visualization

- Critical Alerts
- High Alerts
- Medium Alerts
- Low Alerts





Task 4 – Incident Ticket Creation

Objective

Create a SOC incident ticket

Title

[Critical] Ransomware Detected on Server-X

Description

Indicators detected:

- File: crypto_locker.exe
- IP Address: 192.168.1.50
- Suspicious encryption activity detected

Priority

Critical

Assignee

SOC Analyst

Status

Open

Actions Taken

Endpoint isolated

IOC collection started

Memory acquisition initiated

Task 5 – Escalation Email

Dear Tier 2 Team,

A Critical security alert was detected on the monitored endpoint during routine SOC monitoring. Wazuh generated multiple alerts indicating a possible SSH brute-force attack originating from IP 10.138.226.251 targeting the victim system 10.154.52.251. Indicators of Compromise (IOCs) include repeated failed login attempts, suspicious authentication activity, and abnormal network traffic patterns. Initial containment actions included isolating the affected VM and blocking the attacker IP using CrowdSec. Further investigation and advanced analysis are required to determine the attack scope, persistence mechanisms, and any potential compromise of sensitive data.

Regards,

Sanika Shinde



2. Response Documentation

Objective

Document incident response procedures, investigation activities, and lessons learned.

Tools Used:

- Google Docs
- Draw.io

Task 1 – Incident Response Template

Executive Summary

On 18 August 2025, a phishing email targeted employees in the finance department. One user clicked a malicious link that attempted credential theft. The SOC team detected suspicious login activity and initiated incident response procedures to isolate the affected endpoint and prevent lateral movement.

2.Timeline

Timestamp	Action
2025-08-18 14:00:00	Isolated endpoint
2025-08-18 14:30:00	Collected memory dump
2025-08-18 15:00:00	Reset user credentials
2025-08-18 16:00:00	Blocked malicious domain

3.Impact Analysis

One employee workstation was affected. No evidence of data exfiltration was identified. Temporary disruption occurred during endpoint isolation and credential reset procedures.

4.Remediation Steps:

- Isolated compromised endpoint
- Reset affected user passwords



- Blocked phishing domain and sender
- Updated email filtering policies
- Conducted malware scan on affected systems

5.Lessons Learned

The incident highlighted the importance of user awareness training and rapid detection capabilities. Improvements are needed in phishing simulation exercises, email filtering rules, and incident escalation procedures to reduce response time and minimize organizational risk.

The screenshot shows a Google Docs document titled "Mock Phishing Incident Response". The document content is as follows:

Mock Phishing Incident Response

1.Executive Summary

On 18 August 2025, a phishing email targeted employees in the finance department. One user clicked a malicious link that attempted credential theft. The SOC team detected suspicious login activity and initiated incident response procedures to isolate the affected endpoint and prevent lateral movement.

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- Reset affected user passwords
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- Conducted malware scan on affected systems

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The incident highlighted the importance of user awareness training and rapid detection capabilities. Improvements are needed in phishing simulation exercises, email filtering rules, and incident escalation procedures to reduce response time and minimize organizational risk.

Task 2 – Investigation Steps Log

Log actions for a mock incident:

The screenshot shows a Google Docs document titled "Investigation Log". The document content is as follows:

Investigation Log

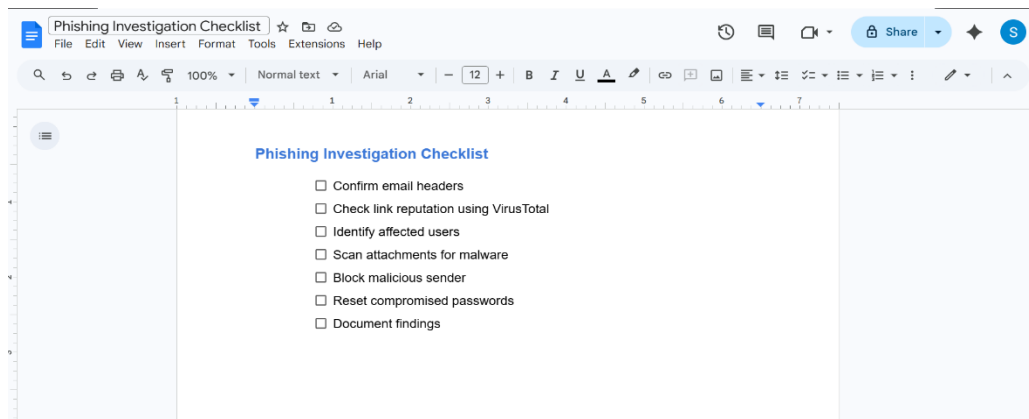
Timestamp	Action
2025-08-18 14:00:00	Isolated endpoint
2025-08-18 14:30:00	Collected memory dump
2025-08-18 15:00:00	Reviewed firewall logs
2025-08-18 15:30:00	Checked suspicious IP reputation
2025-08-18 16:00:00	Closed incident



Task 3 – Phishing Investigation Checklist

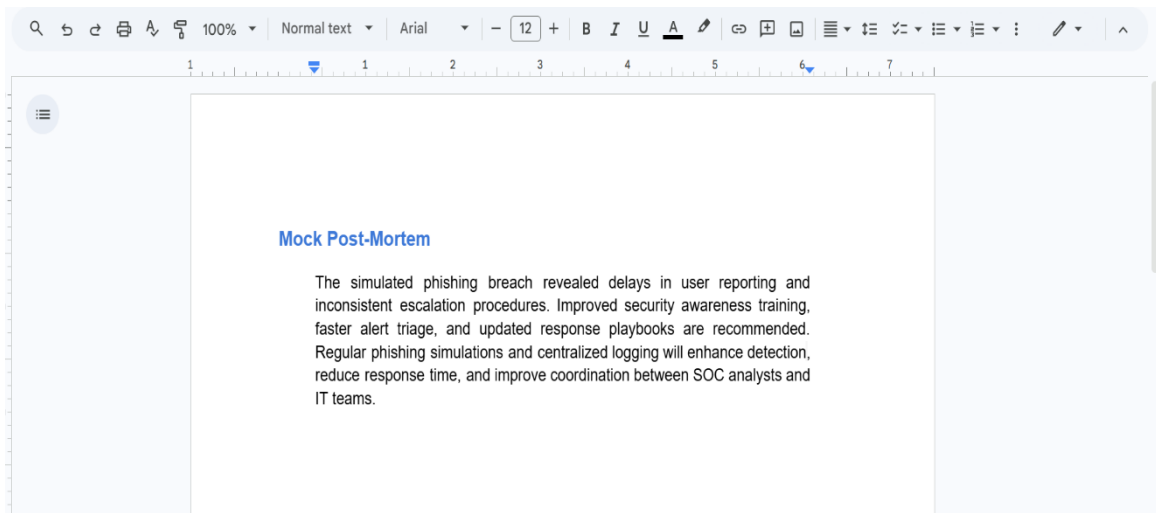
Checklist

- ✓ Confirm email headers
- ✓ Check link reputation using VirusTotal
- ✓ Identify affected users
- ✓ Review email gateway logs
- ✓ Collect endpoint logs
- ✓ Isolate affected endpoint
- ✓ Reset user credentials
- ✓ Document incident timeline



Task 4 – Post-Mortem Summary

The simulated phishing breach revealed delays in user reporting and inconsistent escalation procedures. Improved security awareness training, faster alert triage, and updated response playbooks are recommended. Regular phishing simulations and centralized logging will enhance detection, reduce response time, and improve coordination between SOC analysts and IT teams.



3. Alert Triage Practice

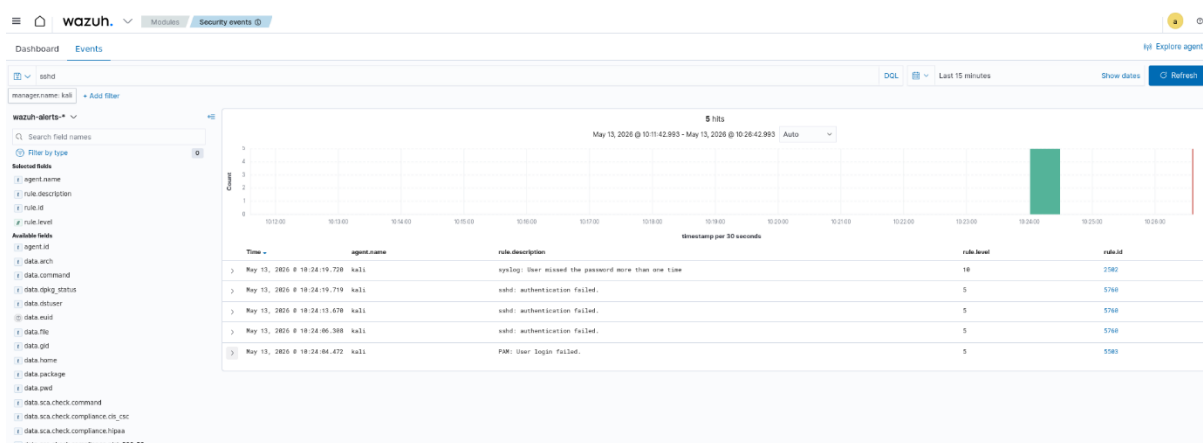
Objective

Analyze alerts, identify false positives, and validate IOCs.

Tools Used

- Wazuh
- VirusTotal
- AlienVault OTX

Task 1 – Triage Simulation





Incident Details

Field	Value
Monitoring Tool	Wazuh
Hostname	kali
Detection Date	May 13, 2026
Event Type	SSH Authentication Failures
Total Alerts	5
Highest Severity	Level 10
Status	Open
Log Source	SSHD / PAM Authentication Logs

Analysis Steps

- Open Wazuh dashboard.
- Review SSH authentication logs.
- Identify repeated failed login attempts.
- Verify source IP reputation.
- Escalate if attack is confirmed.

Triage_Table:

Triage Table										
File Edit View Insert Format Data Tools Extensions Help										
100% 123 Default... 10 B I A										
D28	A	B	C	D	E	F	G	H	I	J
1	Alert ID	Description	Source	Priority	Status					
2	5610	SSH Authentication Failed	10.0.2.15	Medium	Open					
3										
4										
5										
6										
7										
8										
9										
10										



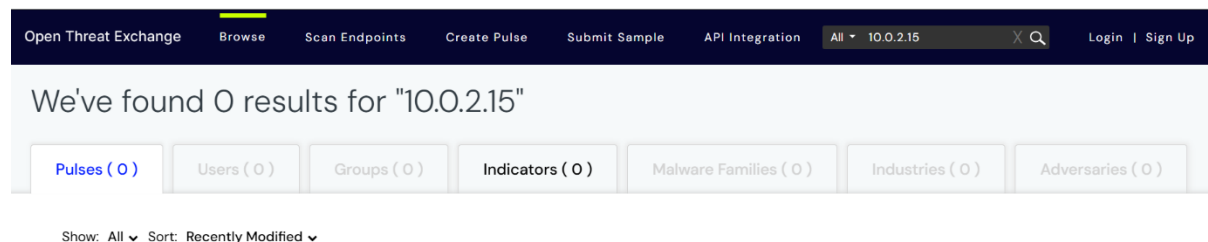
AlienVault OTX

Objective

The objective of this investigation was to validate the suspicious IP address associated with repeated SSH authentication failures using AlienVault OTX threat intelligence services.

Investigation Steps

1. Logged into AlienVault OTX platform.
2. Searched the source IP address:
10.0.2.15



No results found for "10.0.2.15"

Threat Intelligence Findings

Parameter	Observation
Searched IP Address	10.0.2.15
Search Results	No results found
Pulses Detected	0



Parameter	Observation
Indicators Found	0
Malware Associations	None
Threat Intelligence Matches	Not Detected
Risk Assessment	Low

Threat Intelligence Validation:

The Wazuh dashboard detected multiple SSH authentication failure attempts from IP address 10.0.2.15. The alert was investigated using AlienVault OTX for threat intelligence validation. No malicious reputation or indicators were identified because the IP belongs to a private lab environment. The activity was intentionally generated and classified as a false positive.

4. Evidence Preservation

Objective

Practice evidence collection, hashing, and chain-of-custody documentation.

Tools Used

Velociraptor

FTK Imager

Task 1 – Volatile Data Collection

Environment Details

Component	Details
Operating System	Windows 11
Tools Used	Velociraptor, FTK Imager
Analyst Role	SOC Analyst
Collection Type	Volatile Data & Memory Acquisition



Component	Details
Hashing Algorithm	SHA-256

Purpose

Volatile data contains information stored in memory that may be lost after a system reboot or shutdown. Collecting this data helps investigators identify:

- Active network connections
- Suspicious remote sessions
- Malware communication

Procedure Performed

Step 1: Launch Velociraptor

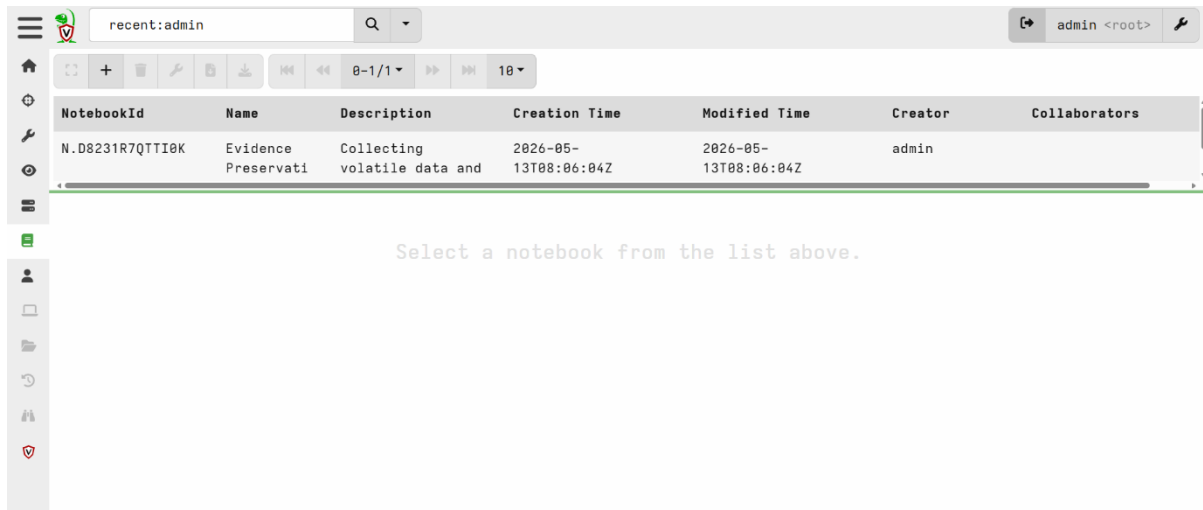
The Velociraptor client was opened on the Windows VM to initiate forensic artifact collection.



Procedure Performed

Step 1: Launch Velociraptor

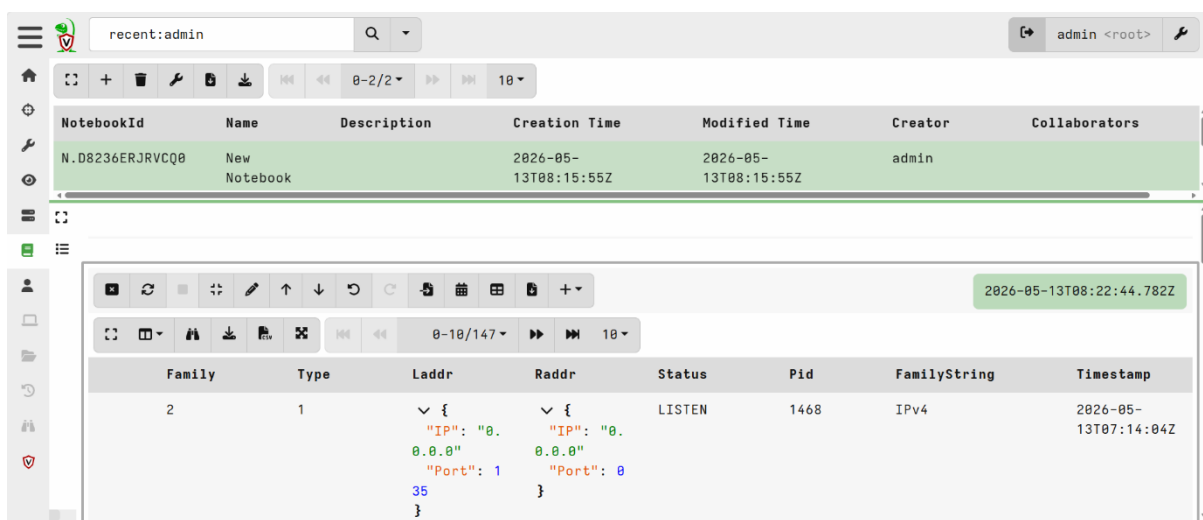
The Velociraptor client was opened on the Windows VM to initiate forensic artifact collection.



Step 2: Execute Netstat Query

The following query was executed to collect active network connection details:

```
SELECT * FROM netstat
```





Step 3: Data Collection

Velociraptor gathered:

- Local IP addresses
- Remote IP addresses
- Connection states
- Protocols (TCP/UDP)
- Listening ports
- Associated processes

Step 4: Export Results

The output was exported and saved in CSV format for later forensic analysis.

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyStr	Timestamp	TypeString
0	2	1	{	{	LISTEN	1468	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	5144	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10648	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10744	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	12908	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	8684	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	3328	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	6156	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP

3. Memory Acquisition

Purpose

Memory acquisition captures the contents of RAM for forensic analysis. This helps identify:

- Running malware
- Injected code
- Active user sessions
- Encryption keys
- Network artifacts
- Malicious processes

Procedure



Step 1: Open Task Manager

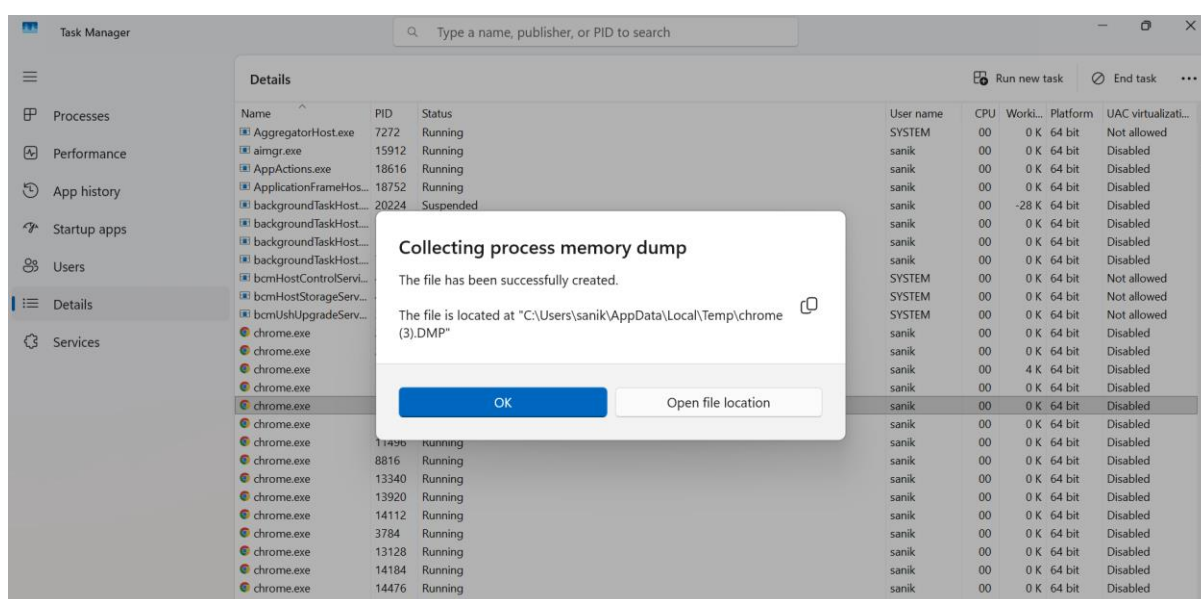
- Press Ctrl + Shift + Esc
- Open the Details tab

Step 2: Select Process

- Identify the target process (example: lsass.exe, explorer.exe, or suspicious process)

Step 3: Create Dump File

- Right-click the process
- Select Create dump file



Step 4: Dump File Location

Windows saves the dump file in:

C:\Users\sanik\AppData\Local\Temp\chrome.DMP

Step 5: Hash the Dump File

Open Command Prompt in the dump location and run:



```
Windows PowerShell (x86) X + v
PS C:\Users\sanik> Get-FileHash "C:\Users\sanik\AppData\Local\Temp\chrome.DMP" -Algorithm SHA256

Algorithm      Hash                                          Path
-----
SHA256         22FD04A44F613349B376B9329B2E637B2DC5B4DDA14346CF27B54153DFC04D13  C:\Users\sanik\AppData\Local\Temp\chrome.DMP

PS C:\Users\sanik> |
```

Chain of Custody Documentation

Purpose

Chain of custody ensures that evidence remains authentic, untampered, and legally admissible.

The screenshot shows a document titled "Evidence Preservation and Chain of Custody" in a word processing application. The document contains a table with the following data:

Item	Description	Collected By	Date	Hash Value
Memory Dump	Chrome Process Dump	Soc Analyst	2026-05-13	22FD04A44F613349B376B9329B2E637B2DC5B4DDA14346CF27B54153DFC04D13
Network Connections	Netstat CSV Export	Soc Analyst	2026-05-13	CSV Export Generated

5. Capstone Project: Full Alert-to-Response Cycle

Objective

The objective of this capstone project was to simulate a real-world cyberattack lifecycle and practice the complete SOC workflow:

1. Attack Simulation
2. Detection



3. Alert Triage
4. Incident Response
5. Containment
6. Reporting and Documentation

Attack Simulation

The attack was performed using the VSFTPD 2.3.4 backdoor exploit available in the Metasploit Framework.

Exploit Used

- Module: exploit/unix/ftp/vsftpd_234_backdoor
- Target Port: 21/TCP
- Vulnerability: VSFTPD Backdoor Command Execution
- MITRE ATT&CK Technique: T1190

Attack Steps:

Step1 — Launch Metasploit

Msfconsole

Step 2 — Select Exploit

use exploit/unix/ftp/vsftpd_234_backdoor

Step 3 — Configure Target

set RHOSTS 10.138.226.251

Step 4 — Execute Exploit

Run

24



Detection and Triage

1	Timestamp	Source IP	Alert Description	MITRE Technique
3	2026-05-14 16:05:12	10.138.226.251	VSFTPD 2.3.4 backdoor exploitation attempt detected	100002
4	2026-05-14 16:06:03	10.138.226.251	Successful exploitation of VSFTPD backdoor	100003
5	2026-05-14 16:06:15	10.138.226.251	Suspicious command execution via FTP	10004

Indicators of Compromise (IOCs)

IOC Type	Value
Attacker IP	10.138.226.251
Target Service	FTP
Target Port	21
Exploit	VSFTPD 2.3.4 Backdoor
Activity	Remote shell access

Incident Analysis

During analysis, it was confirmed that the attacker exploited a publicly accessible FTP service vulnerable to the VSFTPD backdoor vulnerability. The exploit enabled unauthorized shell access to the target machine.

Response:

CrowdSec Mitigation

CrowdSec was configured to block the attacker's IP address.

Command Used

```
sudo cscli decisions add --ip 10.138.226.251 --duration 24h --reason "Attacker blocked"
```



```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)~$ sudo su
[sudo] password for kali:
(root@kali)~/home/kali$ sudo cscli decisions add --ip 10.138.226.251 --duration 24h --reason "Attacker blocked"
INFO Decision successfully added
(root@kali)~/home/kali$ cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
2	cscli	Ip:10.138.226.251	Attacker blocked	ban			1	23h59m38s	2

Block IP

```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)~$ sudo su
[sudo] password for kali:
(root@kali)~/home/kali$ ping 10.138.226.251
PING 10.138.226.251 (10.138.226.251) 56(84) bytes of data.
^C
— 10.138.226.251 ping statistics —
63 packets transmitted, 0 received, 100% packet loss, time 64118ms

(root@kali)~/home/kali$
```

Recommendations

- Disable vulnerable and outdated FTP services.
- Apply security patches regularly.
- Restrict public-facing services using firewalls.
- Enable multi-layer monitoring with SIEM solutions.
- Configure automated IPS blocking for repeated attacks.



Incident Response Report

Date: 14 may 2026

Prepared By: Sanika Shinde

Executive summary:

A simulated cybersecurity incident was conducted in a virtual lab environment using Wazuh, CrowdSec, Kali Linux, and Metasploitable2. The objective was to identify malicious activity, isolate the affected system, and block the attacker IP address. Wazuh generated alerts related to suspicious network activity and possible brute-force attempts. The incident response process was performed successfully.

Timeline

- Attacker machine launched suspicious traffic toward the victim VM.
- Wazuh detected and generated security alerts.
- Logs and alerts were reviewed in the Wazuh dashboard.
- The victim VM network adapter was disabled to isolate the machine.
- CrowdSec was configured to block the attacker IP address.
- Connectivity tests confirmed that communication between attacker and victim was blocked.

Recommendations

- Enable continuous monitoring using Wazuh.
- Regularly update detection rules and signatures.
- Use CrowdSec for automated threat blocking.
- Perform regular vulnerability scans and patch management.
- Maintain incident response documentation and backup procedures.



Stakeholder Briefing:

A cybersecurity incident simulation was completed successfully in the virtual lab environment. Security alerts generated by Wazuh identified suspicious activity originating from the attacker system. Immediate response actions were taken, including isolating the affected virtual machine and blocking the attacker IP address using CrowdSec. Verification tests confirmed that malicious communication was stopped successfully. No permanent damage or data loss occurred during the exercise. The activity demonstrated the effectiveness of monitoring, alerting, and containment procedures. Recommendations include continuous monitoring, regular updates, vulnerability scanning, and improved incident response readiness to strengthen the organization's overall cybersecurity posture.



CYART

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